

STOCKTON, CA, USA

WMO: 724920

Lat: 37.889N

Lon: 121.226W

Elev: 8

StdP: 101.23

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23237

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-0.7	0.6	-5.7	2.3	10.6	-3.7	2.8	9.3	11.9	12.5	10.8	11.8	1.3	130	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	38.6	21.0	36.8	20.5	35.1	20.0	22.8	35.6	21.6	34.6	20.6	33.3	4.5	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.0	29.2	16.5	11.7	26.7	15.5	11.0	25.4	67.2	35.6	62.5	34.7	59.2	33.3	31.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	8.6	7.7	DB	-2.9	41.7	1.8	1.4	-4.2	42.7	-5.2	43.5	-6.2	44.3	-7.5	45.3	(4)
(5)			WB	-3.6	24.7	1.9	3.1	-5.0	26.9	-6.1	28.7	-7.2	30.4	-8.6	32.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.9	8.7	10.7	13.2	15.6	19.3	23.0	25.2	24.7	22.8	18.0	12.3	8.6	(6)	
	DBStd	6.61	2.72	2.53	2.85	3.27	3.37	3.34	2.85	2.57	2.82	2.86	3.03	2.93	(7)	
	HDD10.0	159	57	20	6	1	0	0	0	0	0	0	12	63	(8)	
	HDD18.3	1321	297	214	160	94	28	3	0	0	2	39	181	302	(9)	
	CDD10.0	2668	18	39	104	168	288	389	472	456	383	249	81	19	(10)	
	CDD18.3	788	0	0	1	11	57	142	214	197	136	30	1	0	(11)	
	CDH23.3	10570	0	1	43	255	881	1930	2812	2473	1671	484	21	0	(12)	
(14) Wind	WSAvg	3.3	2.7	3.1	3.3	3.9	4.2	4.3	3.8	3.6	3.2	2.7	2.4	2.9	(14)	
	PrecAvg	355	70	60	50	26	9	2	1	1	7	19	52	51	(15)	
	PrecMax	677	179	209	165	79	95	17	16	21	63	68	158	135	(16)	
	PrecMin	142	4	1	2	2	0	0	0	0	0	0	1	1	(17)	
	PrecStd	136	50	48	39	20	17	3	3	4	14	19	41	36	(18)	
	DB	19.0	22.3	26.8	32.4	36.8	40.0	40.8	39.9	38.0	33.7	25.7	18.4	13.6	(19)	
	MCWB	12.7	14.5	16.6	18.4	20.2	21.5	22.1	21.5	20.6	18.6	15.6	13.6	13.6	(20)	
(23) Mean Coincident Wet Bulb Temperatures	DB	16.9	19.8	24.4	28.9	33.6	37.5	38.6	37.7	35.9	30.4	23.3	16.6	16.6	(21)	
	MCWB	12.3	13.6	15.5	16.9	18.9	20.7	21.1	20.7	20.2	17.4	14.8	12.4	12.4	(22)	
	DB	15.2	17.9	22.5	26.4	31.1	35.1	36.6	35.8	34.0	28.2	20.8	15.1	15.1	(23)	
	MCWB	11.6	12.5	14.5	15.7	17.6	19.5	20.6	20.1	19.3	16.2	13.9	11.8	11.8	(24)	
	DB	13.6	16.2	20.3	23.8	28.4	32.6	34.7	33.7	31.8	26.0	18.6	13.8	13.8	(25)	
	MCWB	10.9	11.9	13.6	14.7	16.7	18.7	20.3	19.6	18.6	15.6	12.7	10.9	10.9	(26)	
	WB	15.1	16.1	18.0	19.6	22.0	23.1	26.0	24.0	25.8	19.9	17.4	15.4	15.4	(27)	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	17.0	19.7	24.7	28.5	32.6	37.9	35.5	36.1	33.5	30.5	23.3	17.4	17.4	(28)	
	WB	13.4	14.8	16.5	17.9	19.8	21.5	22.8	21.9	21.5	18.4	15.8	13.9	13.9	(29)	
	MCDB	15.3	18.3	22.9	26.9	31.8	35.2	35.9	35.4	32.4	28.4	21.2	15.5	15.5	(30)	
	WB	12.2	13.6	15.3	16.5	18.4	20.4	21.4	20.8	20.0	17.4	14.8	12.7	12.7	(31)	
	MCDB	14.1	16.9	20.9	24.9	29.6	33.1	34.8	33.6	31.4	26.1	19.2	14.1	14.1	(32)	
	WB	11.3	12.5	14.1	15.3	17.2	19.3	20.5	20.0	19.0	16.3	13.8	11.5	11.5	(33)	
	MCDB	13.1	15.5	19.2	22.3	27.1	31.3	33.5	32.3	30.4	24.2	17.6	13.1	13.1	(34)	
(35) Mean Daily Temperature Range	MDBR	8.9	10.8	12.6	13.9	15.5	17.3	18.2	17.9	17.2	15.3	12.0	9.4	9.4	(35)	
	5% DB	11.8	13.9	16.3	18.7	19.9	20.6	20.3	20.1	20.2	18.9	15.5	11.1	11.1	(36)	
	MCWBR	7.5	8.2	8.5	8.4	8.1	7.8	7.7	7.5	8.0	8.2	8.3	7.1	7.1	(37)	
	5% WB	8.5	11.7	14.6	17.0	18.7	19.2	19.2	18.7	18.2	16.7	12.9	7.1	7.1	(38)	
	MCWBR	5.8	7.3	8.1	8.3	8.5	7.8	8.4	7.7	8.2	7.9	7.4	4.9	4.9	(39)	
(40) Clear-Sky Solar Irradiance	taub	0.333	0.331	0.343	0.362	0.365	0.362	0.368	0.352	0.355	0.353	0.342	0.332	0.332	(40)	
	taud	2.469	2.486	2.439	2.380	2.391	2.411	2.402	2.431	2.425	2.439	2.445	2.478	2.478	(41)	
	Ebn at Noon	846	900	920	917	915	914	906	915	894	860	826	820	820	(42)	
	Edh at Noon	82	91	105	118	119	117	117	111	105	94	83	77	77	(43)	
(44) All-Sky Solar Radiation	RadAvg	2.30	3.31	4.72	6.16	7.42	8.31	8.20	7.38	6.02	4.31	2.83	2.07	2.07	(44)	
	RadStd	0.37	0.41	0.47	0.54	0.43	0.31	0.22	0.20	0.20	0.26	0.18	0.32	0.32	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.62	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)		+0.40	N/A	-1.12	N/A	N/A	N/A	N/A	-117	+133	+30

Nomenclature: See separate page